

$$C = (1, x, x^2)$$

$$A = [\phi]_C^C = ([\phi(1)]_C | [\phi(x)]_C | [\phi(x^2)]_C) = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

$$p_A(\lambda) = \begin{vmatrix} -\lambda & 2 & 0 \\ 0 & -\lambda & 4 \\ 0 & 0 & 2-\lambda \end{vmatrix} = \lambda^2(2-\lambda) \quad \begin{array}{l} \nearrow \lambda_1 = 0 \text{ alg nás 2} \\ \searrow \lambda_2 = 2 \text{ alg nás 1} \end{array}$$

geom. nás 1 (2LN)

$$\lambda = 0: \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

$\downarrow$
 A_0

dva řetězci

$$\begin{array}{l} v_2 \rightarrow v_1 \rightarrow 0 \\ v_3 \rightarrow 0 \end{array}$$

$$\begin{array}{l} v_1 \in \text{Ker } A_0 \cap \text{Im } A_0 \in \text{span}\left\{\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}\right\} \\ \text{Zvolím } v_1 = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \\ v_2 = A_0 v_1 = \begin{pmatrix} 0 \\ 8 \\ 0 \end{pmatrix} \end{array}$$

$$v_3: \lambda = 2: \begin{pmatrix} -2 & 2 & 0 \\ 0 & -2 & 4 \\ 0 & 0 & 0 \end{pmatrix} v_3 = \vec{0}$$

$$v_3 \in \text{span}\left\{\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}\right\}$$

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$B = (v_1, v_2, v_3) = (2, x, 2+2x+x^2)$$

$$[\phi]_B^B = J$$